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Unlocking Hidden Recovery from Mature Carbonate: Field E Dual Reservoir cGOGD Strategy

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Abstract

This paper presents the development plan of a technically complex and economically rejuvenated oil recovery project in Field E, a mature fractured carbonate field in North Oman. The objective is to unlock contingent resources from the Natih E and Shuaiba reservoirs by implementing cold Gas-Oil Gravity Drainage (cGOGD) and leveraging learnings from a successful analogue. The scope includes optimizing field layout, well count, and surface facility decisions to enable a phased redevelopment strategy with minimal environmental impact and improved economic competitiveness.

An integrated development workflow was used, combining static and dynamic modeling, dual-reservoir simulation, and fracture network characterization. Thirty-three horizontal oil producers supported by gas lift, four crestally placed gas injectors, six observation wells, and aquifer pump-off (APO) wells were strategically planned. Analogue benchmarking from the nearby GOGD project informed the approach to fracture avoidance in injectors, APO sequencing, and gas injection management. Facility design was optimized through scenario screening, evaluating brownfield vs greenfield options and compressor sizing to ensure seamless integration with subsurface plans. Gas Breakout Control (GBC) technologies were incorporated to mitigate short-circuiting risks.

The development plan nearly doubles the recovery factor from both reservoirs compared to historical performance. The project achieved a reduction in unit technical cost (UTC) by integrating gas recycling, fit-for-purpose facility design, and removal of gas sweetening requirements. The economic case is resilient, with a positive NPV and a low payback period. Simulation results confirmed that reservoir pressure support and gas cap evolution are well managed under the proposed GOGD layout. Key risks such as H₂S exposure, early gas breakthrough, and compressor limitations were mitigated through material selection, injector placement, and phased design.

This paper demonstrates how GOGD can be successfully applied to a dual-reservoir, fractured carbonate field to transform a long-stranded asset into a viable, economically robust development. The planning framework integrates analogue learnings, fracture risk management, APO well contingency, and zero-flaring operations. The unique combination of dual-reservoir GOGD, facility reuse optimization, and gas handling strategy offers a replicable model for mature asset revitalization across the region. It provides new

insight into how low-permeability, high-complexity fields can be competitively redeveloped using gravity-driven mechanisms without relying on intensive hydraulic stimulation.

Introduction

Maximizing recovery from mature carbonate reservoirs remains a key priority for upstream operators in the Middle East, where a large share of undeveloped hydrocarbons lies within low-permeability, naturally fractured systems. These reservoirs present significant challenges, such as poor sweep efficiency under waterflooding, early breakthrough of injected fluids, and complex geological architecture—all of which can undermine conventional recovery approaches.

Field E, located in North Oman, exemplifies this complexity. It comprises a stacked system of three fractured carbonate reservoirs: Natih A, Natih E, and Shuaiba. Production began in the 1980s, initially via vertical and later horizontal wells supported by gas lift. Despite decades of intermittent production, cumulative recovery remains below 10% of the original oil in place.

Following comprehensive technical screening, the redevelopment strategy shifted focus to the Natih E and Shuaiba reservoirs, leveraging a cold Gas-Oil Gravity Drainage (cGOGD) mechanism. While the Natih A reservoir holds significant volumes, it was excluded from the current development scope due to its high oil viscosity, low vertical communication, large gas cap, and need for further appraisal.

cGOGD was selected as the optimal recovery mechanism based on its ability to exploit gravity segregation in fracture-dominated settings without introducing the water handling complexity associated with waterflood. The approach builds on successful analogue implementation in a nearby field, where cGOGD yielded strong economic returns with low carbon footprint and minimal operational intensity.

The Field E project was enabled by a confluence of technical advances and strategic learnings:

- Dual-reservoir compositional simulation models that integrate Natih E and Shuaiba responses under unified injection strategies;
- Enhanced fracture network understanding using borehole image logs and seismic data;
- Analogue-derived design philosophies including injector avoidance of fracture corridors, staged gas ramp-up, and use of Aquifer Pump-Off (APO) wells as contingency; and
- A surface concept eliminating gas sweetening and water treatment, supported by compression infrastructure tailored to match ramp-up and sour gas handling.

In total, the project targets to unlock contingent oil resources—from Natih E and Shuaiba—achieving a lower Unit Technical Cost (UTC) and positive NPV. Recovery factors are expected to more than double to ~17–18% for each reservoir—all while maintaining zero routine flaring.

This paper outlines the integrated full-field redevelopment strategy of Field E. It covers the subsurface development plan, dynamic simulation results, surface facility alignment, and phased execution strategy that brought the project to assurance readiness. The Field E cGOGD case presents a replicable blueprint for revitalizing complex, fractured brownfields across the region in an economically and environmentally sustainable way.

Field Overview and Development Opportunity

Field E, located in North Oman, consists of a structurally complex stack of Cretaceous-age, naturally fractured carbonate reservoirs. Discovered in the early 1970s and brought on stream in 1985, the field has produced intermittently for over 35 years. However, due to its challenging reservoir characteristics and surface facility constraints, cumulative recovery has remained low (see [Figure 1](#) for field location).

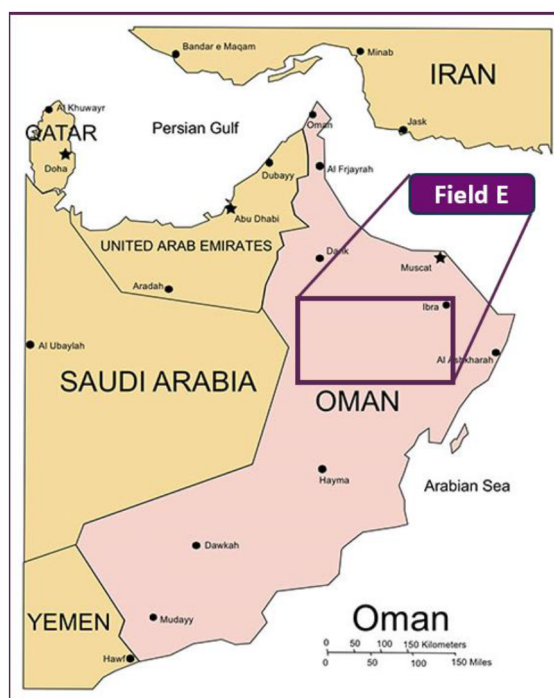


Figure 1—Field E Location Map

Reservoir Architecture

The field contains three main reservoir intervals:

- **Natih A:** While volumetrically significant, this unit is excluded from the current redevelopment plan due to its high oil viscosity, low vertical connectivity, and extensive gas cap. It has been deferred for future appraisal.
- **Natih E:** A ~105 m thick carbonate unit showing a shallowing-upward sequence with favorable matrix porosity in the upper layers. It exhibits moderate support from an underlying aquifer and hosts discrete fracture corridors.
- **Shuaiba:** A thinner (~35 m) reservoir composed of laterally extensive shallow marine carbonates. The top leached rudist layer demonstrates very high permeability (up to 1.5 Darcy) and is critical to the success of gravity drainage.

Historical recovery factors remain low—approximately 9.3% for Natih E and 10.0% for Shuaiba—leaving 35% of recoverable contingent resources under a revitalized recovery strategy (refer to Table 1 for reservoir property comparison).

Table 1—Key Subsurface Data for Field E Reservoirs

Property	NatihE	Shuaiba
Reservoir Lithology	Limestone	Limestone
Max Oil Column -m	75	35
Porosity Range -%	13 - 39	16 - 39
Matrix Permeability -Md	0.1 - 70	1 - 23
API Gravity	20.3 - 20.8	26.6
Oil Viscosity (@res. cond.) -cp	32	8.17
Initial Reservoir Pressure -Kpa	4,964	7,220

Property	NatihE	Shuaiba
Bubble Point Pressure -Kpa	4,909	6,582
Current Reservoir Pressure -Kpa	3,700	6,240
Reservoir Temperature -C	47	64
H ₂ S Content in Gas -%	4.4	4.4

Figure 2 presents a generalized geological cross-section illustrating the relative positions of the Natih A, E, and Shuaiba accumulations within the structure.

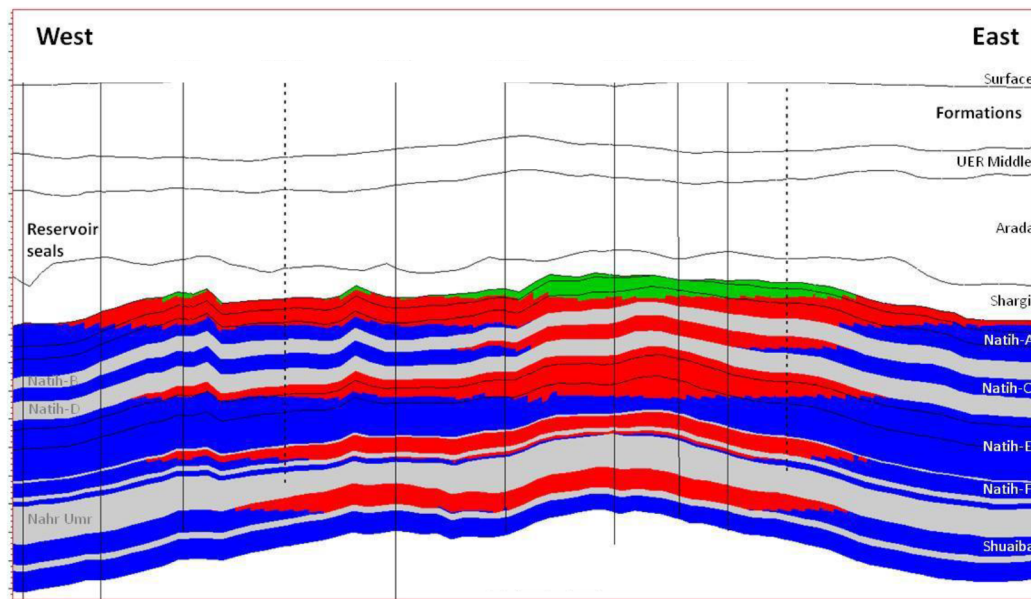


Figure 2—Generalized geological cross section showing the different hydrocarbon accumulations within field E

Historical Development

Initial development involved vertical wells targeting both Natih E and Shuaiba. Later, horizontal wells with gas lift were introduced to improve productivity. While these helped slow production decline, challenges related to poor matrix-fracture connectivity and early water/gas breakthrough persisted.

Over the past two decades, several redevelopment concepts were evaluated:

- Waterflooding in Natih E,
- Dual-lateral horizontal producers,
- Gas re-injection for pressure maintenance, and
- Hydraulic fracturing trials.

However, each was deemed technically or economically unviable. For instance, a 2015-2016 evaluation of a combined Natih E-Shuaiba waterflood and GOGD strategy showed a high UTC—exceeding economic thresholds. Consequently, development was paused until 2019, when a revised concept centered around cold Gas-Oil Gravity Drainage (cGOGD) was introduced.

Opportunity Reframing

The adoption of cGOGD as a redevelopment strategy marked a fundamental shift in the field's recovery philosophy. Key enablers of this transition included:

- Favorable fracture geometry and matrix properties conducive to gravity segregation;
- Strong analogue performance in nearby fields, demonstrating the viability of sour gas reinjection without sweetening;
- Zero flaring strategy, contributing to PDO's decarbonization roadmap;
- Surface facility optimization, removing high-cost elements like gas sweetening and desalting;
- Use of **Aquifer Pump-Off (APO)** wells as backup for pressure support;
- Phased development with fit-for-purpose compressor sizing and modularity.

The approved development scope includes:

- **33** horizontal oil producers (with gas lift),
- **4** crestally placed gas injectors,
- **6** observation wells, and
- **Up to 6** APO wells depending on aquifer performance.

This strategy unlocks 35% of recoverable oil—from Natih E and Shuaiba—while reducing technical cost and facility complexity.

Subsurface Development and Simulation Approach

The Field E redevelopment was guided by a technically rigorous subsurface strategy aimed at unlocking stranded oil through gravity-driven recovery mechanisms, while mitigating the risks posed by complex fracture networks, high H₂S content, and significant depletion. An extensive evaluation of static and dynamic uncertainties was performed to arrive at a development plan that is technically robust, economically resilient, and operationally feasible.

Development Concept and Philosophy

The core concept leverages cold Gas-Oil Gravity Drainage (cGOGD) to enable downward drainage from crestally injected gas, displacing oil toward horizontal producers placed at mid-flank elevations. The strategy minimizes drawdown, controls coning, and allows for full gas recycling without surface sweetening.

Key pillars of the development philosophy include:

- Dual-reservoir development: Natih E and Shuaiba were jointly modeled but optimized independently to accommodate structural differences and fluid properties.
- Gas injection from crest: Designed to ensure efficient vertical sweep while managing fracture connectivity risks.
- Horizontal production at mid-flank: Producer placement maximizes gravity segregation and delays gas breakthrough.
- Retention of APOs: Strategic use of Aquifer Pump-Off wells provides optionality for pressure support or water removal if required.

As shown in [Figure 3](#), the schematic illustrates the producers, gas injectors, and APO well configuration in the cGOGD recovery process.

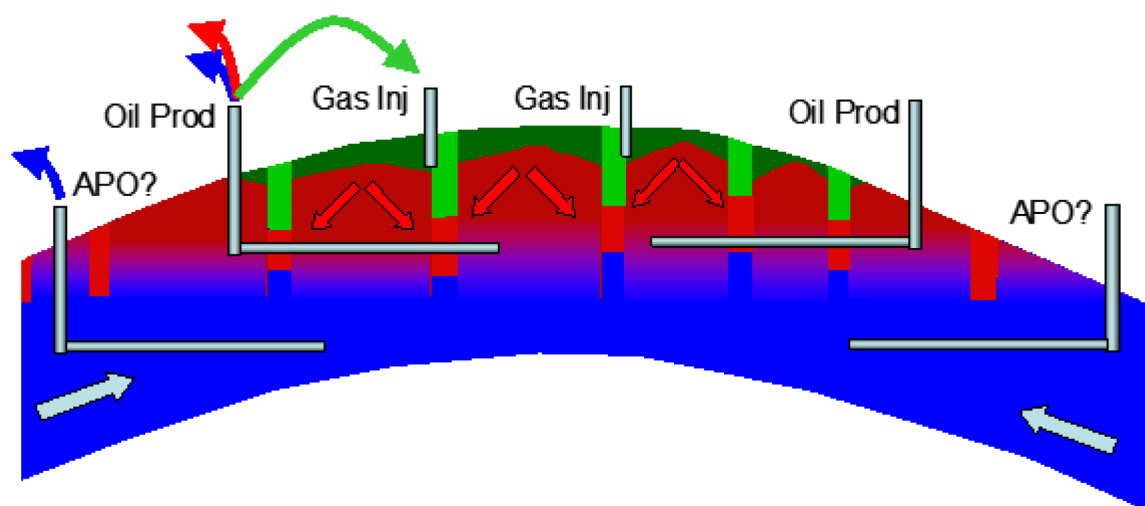


Figure 3—Schematic showing the producers, Gas Injector and APO well configuration in GOGD recovery process.

Static Model and Fracture Characterization

Both reservoirs are naturally fractured carbonates. Fracture identification was based on FMI logs, core data, and analogues. A Discrete Fracture Network (DFN) was constructed and stochastically upscaled into dual-porosity, dual-permeability static models.

Seven fracture realizations were tested dynamically to assess:

- Sensitivity to fracture intensity and anisotropy.
- Short-circuiting risks from crest to flanks.
- Impact on GOC movement, vertical conformance, and plateau duration.

These realizations were key to defining perforation strategies and gas injector depths (refer to Figure 4).

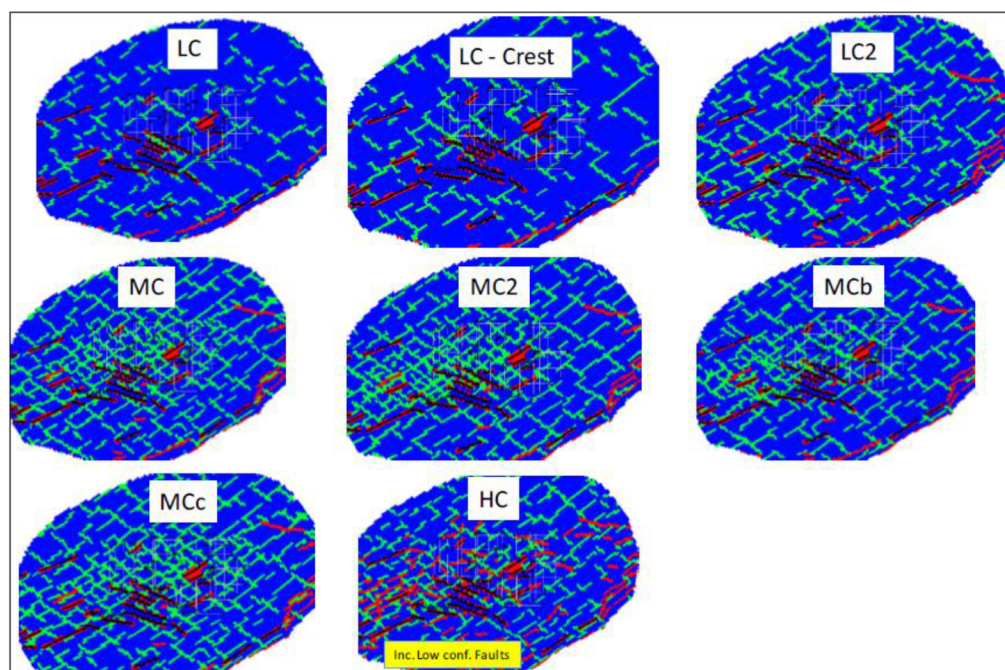


Figure 4—Seven fracture model cases considered for history matching and prediction.

Model Initialization. The dynamic model was initialized with Pc curves derived from the saturation height function. Checks were performed for volumetrics, saturation profiles, and HCPV. Results are shown in Figure 5, comparing dynamic model initialization against the static model.

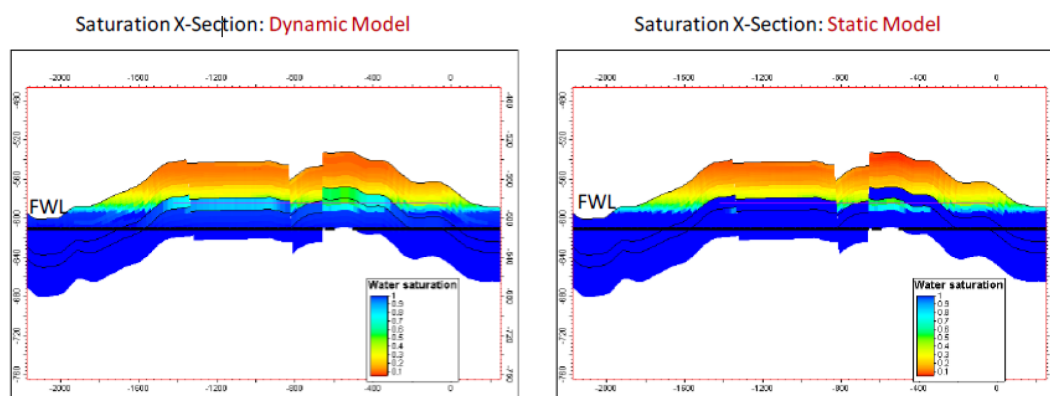


Figure 5—Initialization check of saturation compared with static model

Sensitivity Analysis. A One Parameter At A Time (OPAAT) approach and deterministic simulations were used to test the impact of fracture parameters, leached zone scenarios, and relative permeability.

Tornado plots in Figure 6 show key sensitivities for both primary production and GOGD recovery:

- Primary recovery: Influenced by fracture permeability, fracture corridor intensity, and aquifer strength.
- GOGD recovery: Driven by vertical matrix permeability, leached zone conductivity, and fracture spacing.

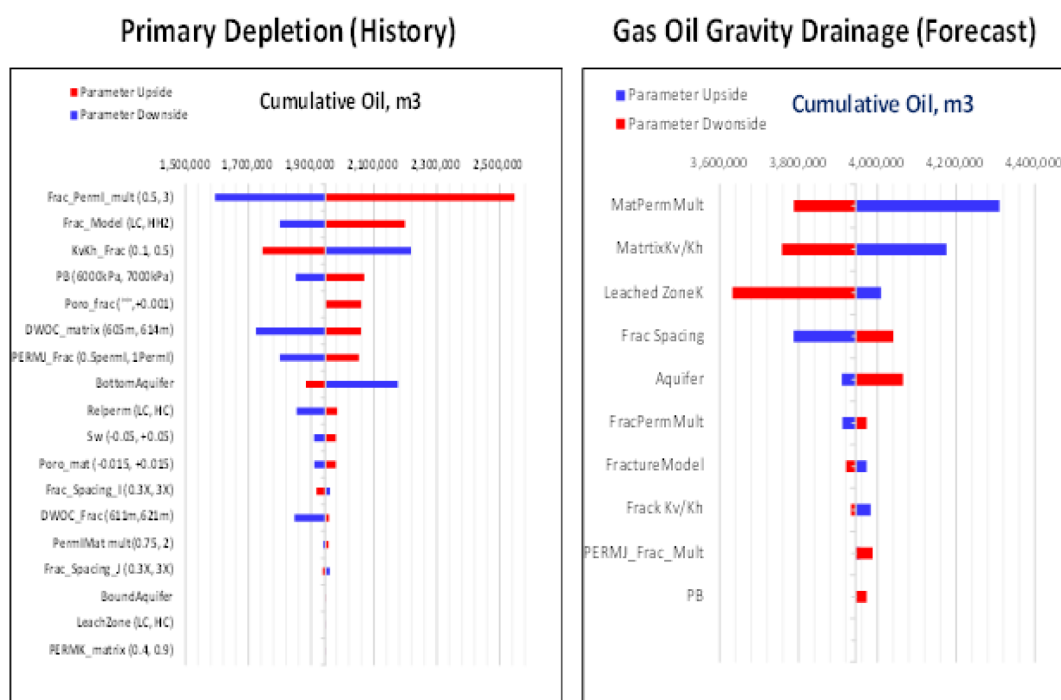


Figure 6—Cumulative oil production sensitive to the key parameters under Primary depletion (Left) and GOGD recovery (Right).

History match Approach. History matching was carried out using both deterministic and assisted techniques within CMG-CMOST to constrain reservoir uncertainties through alignment with historical performance data. The primary objective was to calibrate the models to support future predictions, recognizing that the change in recovery mechanism—from primary production to cold GOGD—limits the ability to fully constrain all uncertainties through historical data alone.

Over 150 dynamic realizations were generated, using a combination of Latin Hypercube Design of Experiments (DoE) and deterministic perturbations on key parameters including fracture intensity, relative permeability, and leached zone characteristics. Around 50 realizations were selected based on their ability to match historical cumulative oil production and average reservoir pressure within ± 10 -20% margins (see Figure 7).

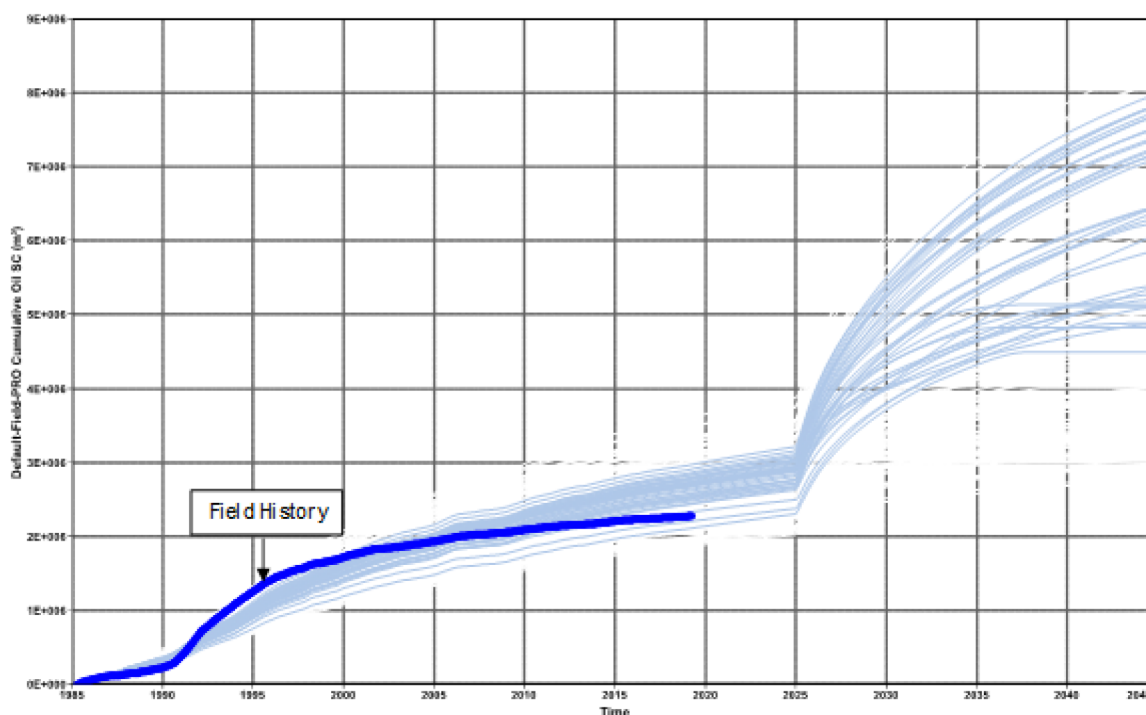


Figure 7—Dynamic realizations from DoE. The blue colour data points are historical cumulative oil production.

The base forecast scenario assumed 15 oil producers and 3 gas injectors, with maximum liquid rates of 400 m³/day per well and continuous gas injection at 150,000 m³/day. A maximum GOR limit of 1000 v/v was enforced, triggering automatic rate cutbacks when exceeded. The history-matched ensemble was used to assess forecast robustness and analyze key drivers for both primary recovery and GOGD performance.

Key Subsurface decisions

Injector Optimization. Injector design focused on balancing gravity drainage efficiency with containment and conformance. The following options were tested:

- Location: Injectors placed 10-15 m below crest to avoid override and enable downward drive.
- Rate: 80 to 150 km³/d tested per injector. 100-120 km³/d selected to balance performance and compression constraints.
- Perforation Depth: Zones away from high fracture corridors were selected to avoid early breakthrough.
- Number of Injectors: Two per reservoir (Shuaiba & Natih E), totaling four.

Simulation showed that increased injection rate beyond 120 km³/d offered minimal incremental recovery and introduced GOR risks as shown in Figure 8.

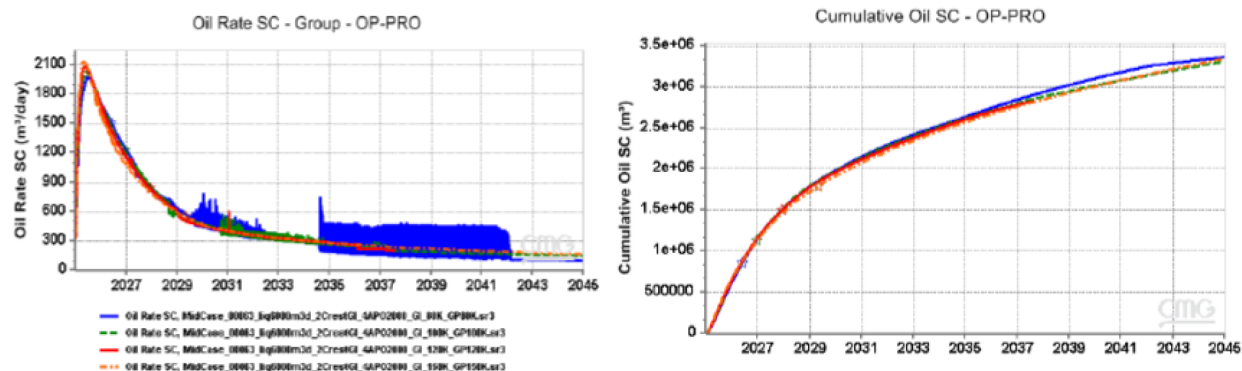


Figure 8—Oil recovery rate and cumulative for 4 gas injection cases

Producer Placement and Depth Sensitivity. Thirty-three horizontal producers were planned along mid-flank to lower flank elevations. Sensitivity studies were conducted on depth, spacing, and trajectory. Key findings:

- Optimum Depth: 385 mTVDss for Natih E, 585 mTVDss for Shuaiba. Shallower wells reduced water cut but sacrificed oil volume.
- Spacing: 300 m spacing balanced cost and vertical sweep.
- Orientation: E-W wells intersect N-S fractures, maximizing lateral connectivity.
- Short Perforations: Used to delay gas breakthrough and manage drawdown.

Figures 9–11 summarize well placement, oil rate outcomes, and cumulative recovery for six simulated scenarios in NatihE.

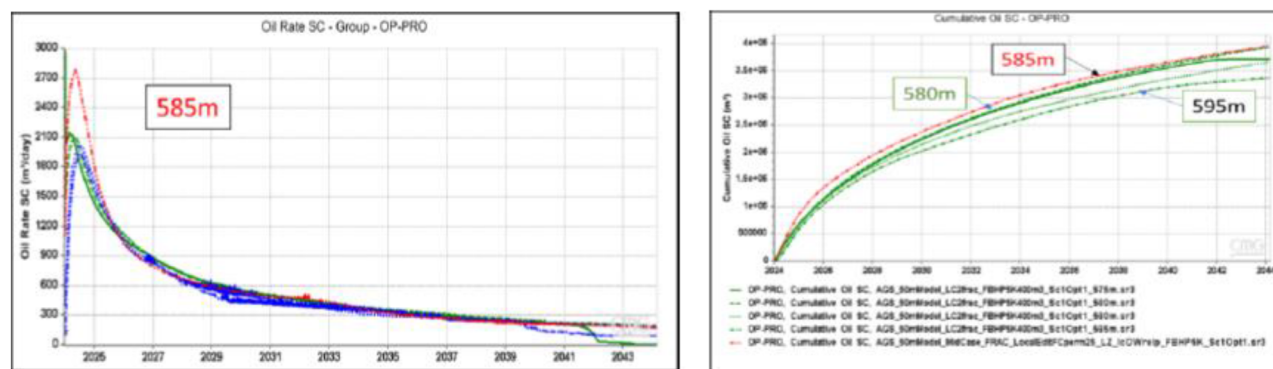


Figure 9—Oil rates and cumulative oil recovery plots for horizontal well placement depth 475m - 495m

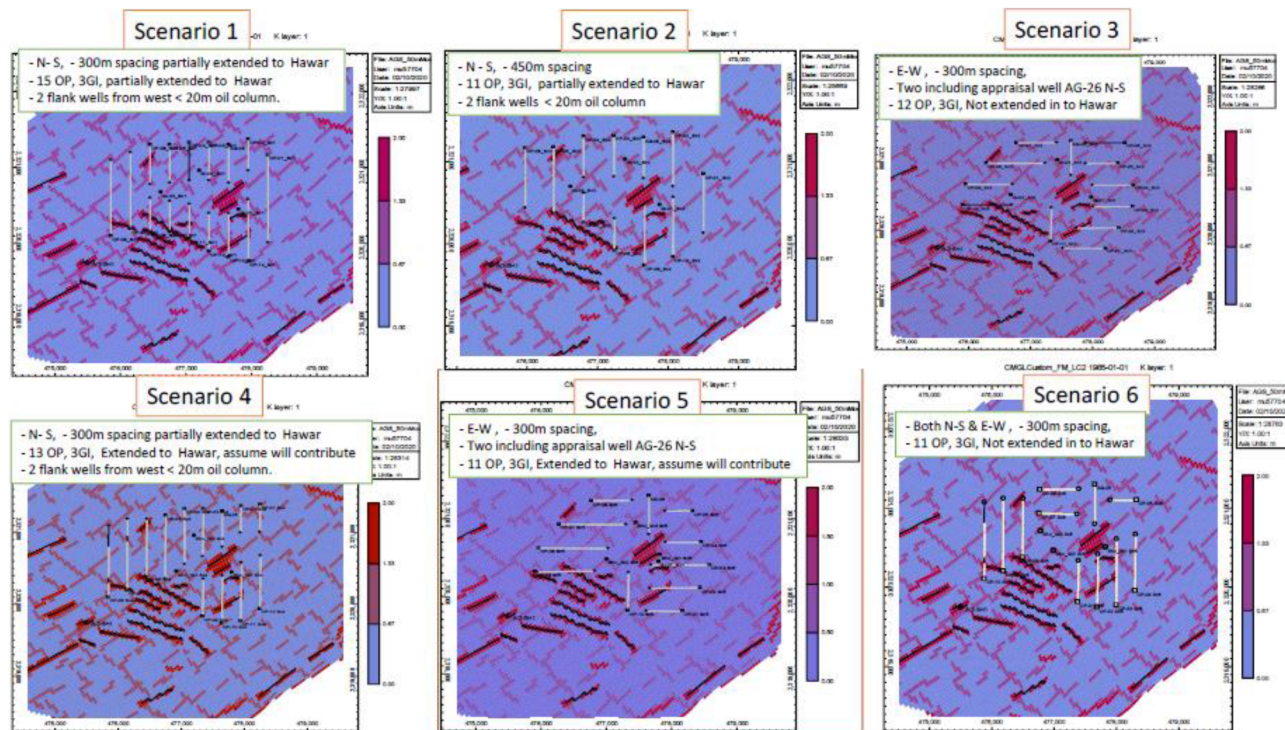


Figure 10—Location map of six oil producers scenarios simulated in dynamic model.

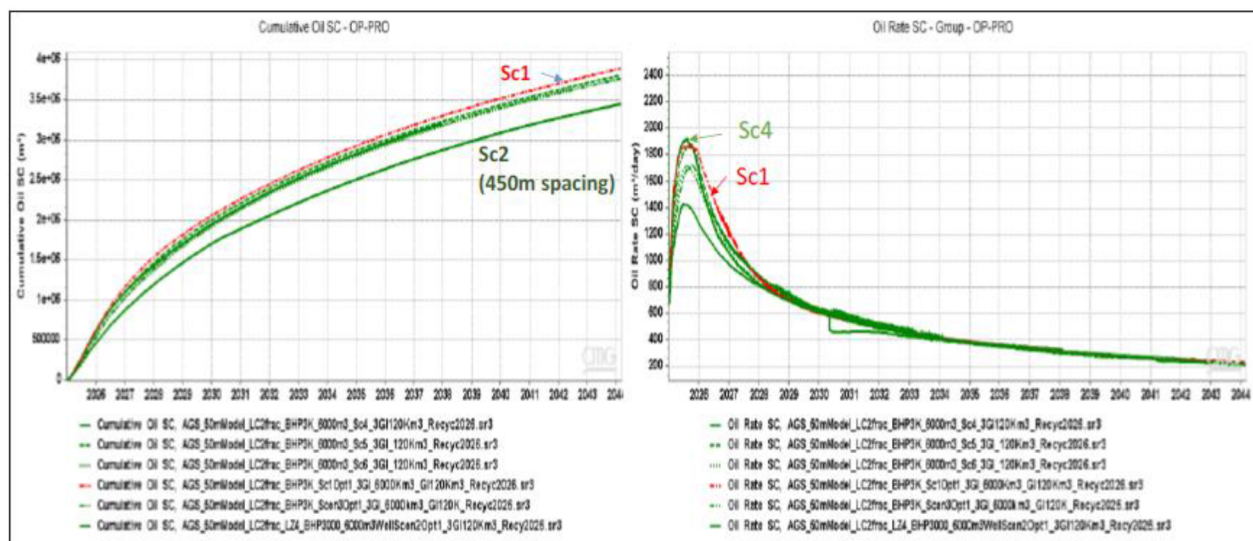


Figure 11—Cumulative oil recovery (Left) and oil rate (Right) prediction for 6 well scenarios. The scenario is having highest cumulative oil recovery and maximum oil rate peak.

APO Strategy. Although no significant aquifer activity is observed, 2-4 APO wells per reservoir were retained to manage pressure imbalance or unexpected water ingress. APO locations were selected in flank/water-leg transition zones. Their activation remains a future contingency (refer to Figure 12)

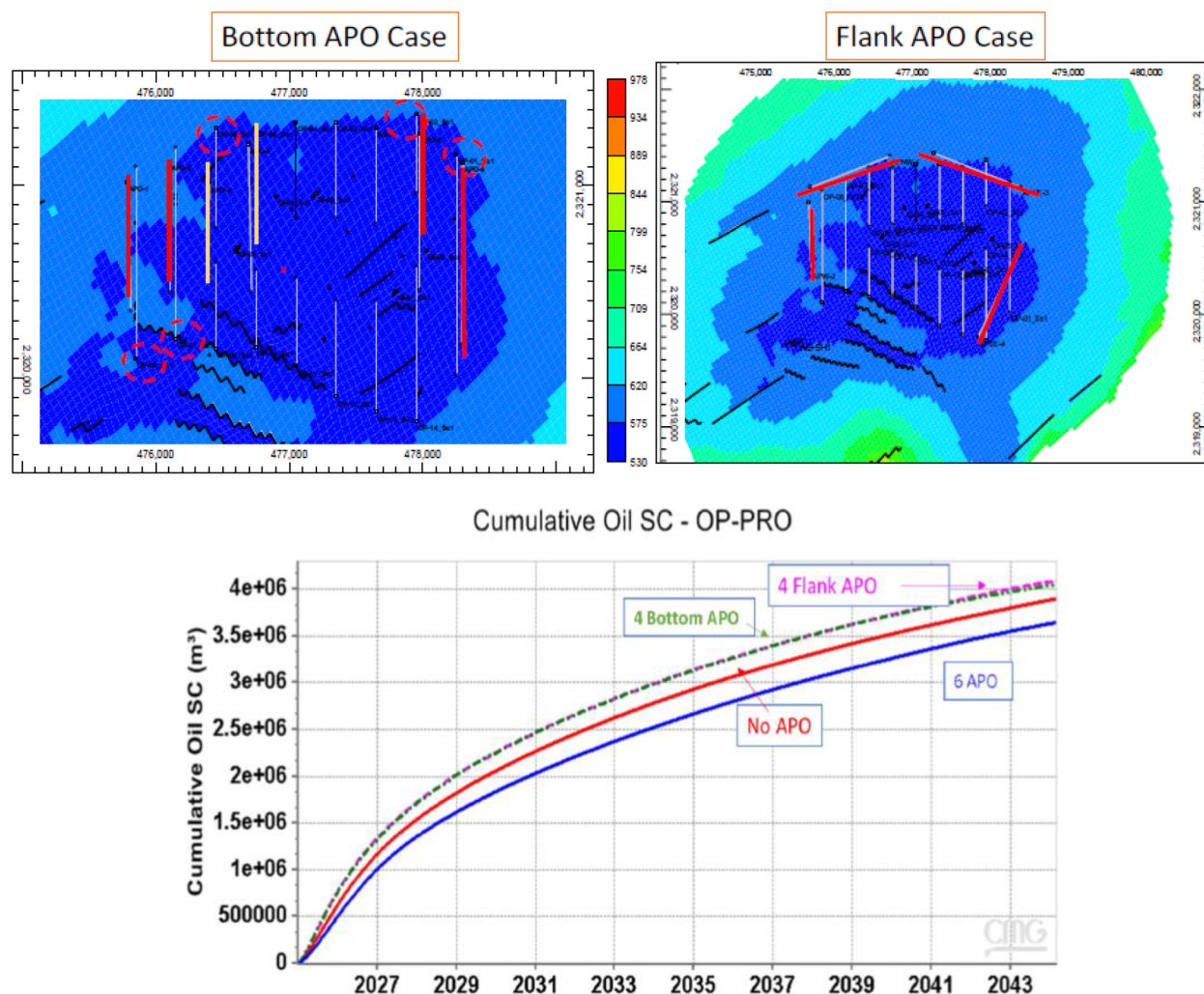


Figure 12—APO well scenarios studied

Dynamic Model and Simulation Results

A full-field dual-reservoir model was developed in INTERSECT, calibrated with historical pressure, GOR, and BHP data. Key features include:

- Fine layering (1 m) near crest to resolve GOC evolution.
- Fluid systems include H₂S, with EOS-tuned PVT.
- GOGD behavior explicitly modeled through gravity override physics.

Base case simulation showed:

- Recovery uplift: From <9% to ~18%.
- GOR control: Maintained <200 m³/m³ through phased injector management.
- Minimal water production under base assumptions.

Table 2—Summary of Development Plan for Natih E and Shuaiba Reservoirs

Well type	Element	Natih E	Shuaiba
Oil Producers	Well Spacing (m)	300	300
	Well number	15	15
	Well Depth (inTVDss)	375	585
	Well Orientation	N-S	N-S
	Well Liquid Rates (m3/d)	200	300-600
Gas Injectors	Well Number	2	2
	Gas Injection Rate (km3/d)	100	120
	Gas Injection Strategy	Recycled GI with No make-up gas	Recycled GI initially with make-up gas from SOGL to reach target rate
APO Wells (Contingent)	Well Number	2	4
	Well Location	Selected based on field performance	Selected based on field performance
	Well Rates (m3Zd)	500	500
Observation Wells	Well Number	3 (2 conversion, 1 new drill)	3 (2 conversion, 1 new drill)

Facilities Design and Optimization

To support the cGOGD development, three surface concept options were evaluated:

- Greenfield development with new dedicated facilities,
- Brownfield reuse leveraging existing infrastructure in nearby clusters, and
- Hybrid concepts combining new and existing systems.

After screening technical, economic, and environmental trade-offs, the selected concept adopted a fit-for-purpose greenfield facility, optimized for:

- No gas sweetening or desalting — made possible by reinjecting sour gas and achieving sufficient H₂S dilution through oil production streams,
- Full gas reinjection with zero routine flaring, reducing GHG intensity to 0.077 tCO₂/tHC, and
- Phased compression aligned with production ramp-up to defer capital spend.

The facility was designed with the following core capacities and features:

- Oil handling capacity: 2,000 m³/d,
- Gas injection capacity: 120,000 m³/d,
- Gas lift (GL) and gas injection (GI) headers,
- Well test header and production metering systems,
- RMS (Remote Manifold Station) and utility packages,
- Hook-ups to export pipelines for oil evacuation and flare lines for safety.

This surface concept was tightly integrated with the subsurface development plan, ensuring facility sizing and operability matched the simulated production and injection forecasts. Key surface elements were also

de-risked through analogy with nearby GOGD facilities. Figure 13 illustrates the schematic layout of the proposed facility, showing the major components supporting GOGD operation.

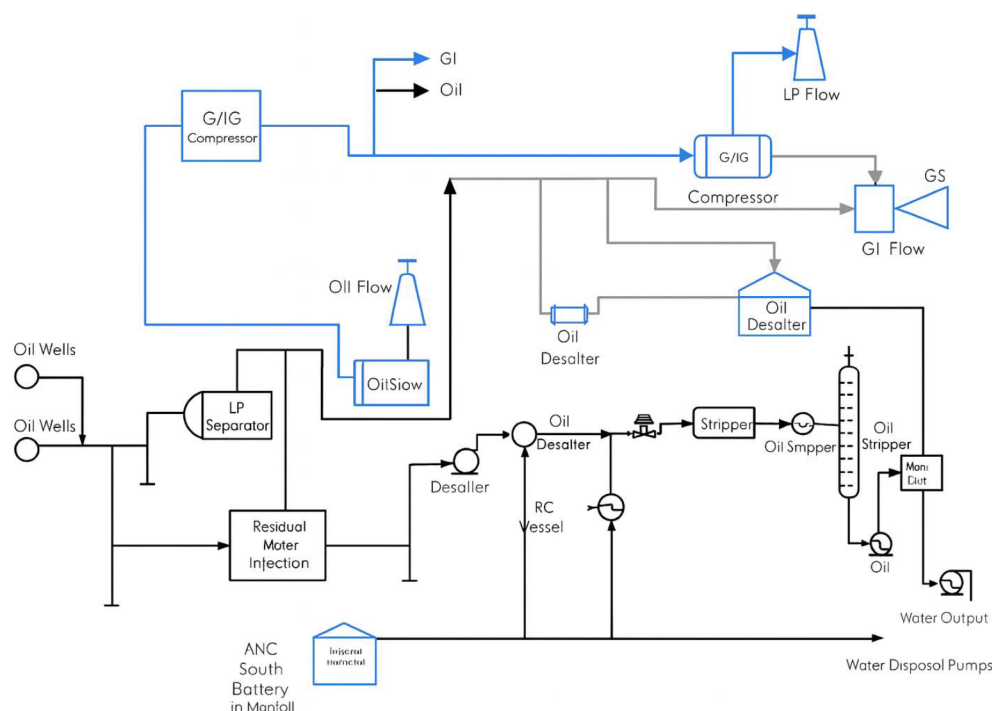


Figure 13—Field E Facility Schematic

Results and Economic Justification

The selected Field E cGOGD development plan delivers significant improvement in both technical recovery and economic performance, transforming the project from a previously marginal opportunity into a robust and highly attractive redevelopment. This section presents the key recovery results, economic outcomes, and supporting sensitivities that underpin the project's viability.

Recovery Factor Uplift

The implementation of cold gas injection through a gravity-assisted GOGD mechanism has unlocked considerable additional recoverable volumes from both Natih E and Shuaiba reservoirs. The process exploits gravity segregation to improve vertical sweep efficiency—particularly beneficial in fractured, low-permeability settings where conventional waterflooding or depletion would be ineffective.

Table 3—Incremental recovery factor gains from cGOGD implementation

Reservoir	Historical RF	Post-GOGD RF	Incremental Recovery
Natih E	~9.3%	~17%	+7.7%
Shuaiba	~10%	~18.2%	+8.2%

The two reservoirs combined forecast to deliver **35%** of incremental recoverable oil, previously stranded due to low reservoir energy and high water cut risk under conventional development.

Key technical outcomes from dynamic simulation include:

- Sustained reservoir pressure support over 20+ years,

- Controlled GOC evolution with delayed gas breakthrough,
- Zero water production under the base-case realization.

Figure 14 presents the oil recovery profiles for low, base, and high cases, confirming the robustness of the base development concept.

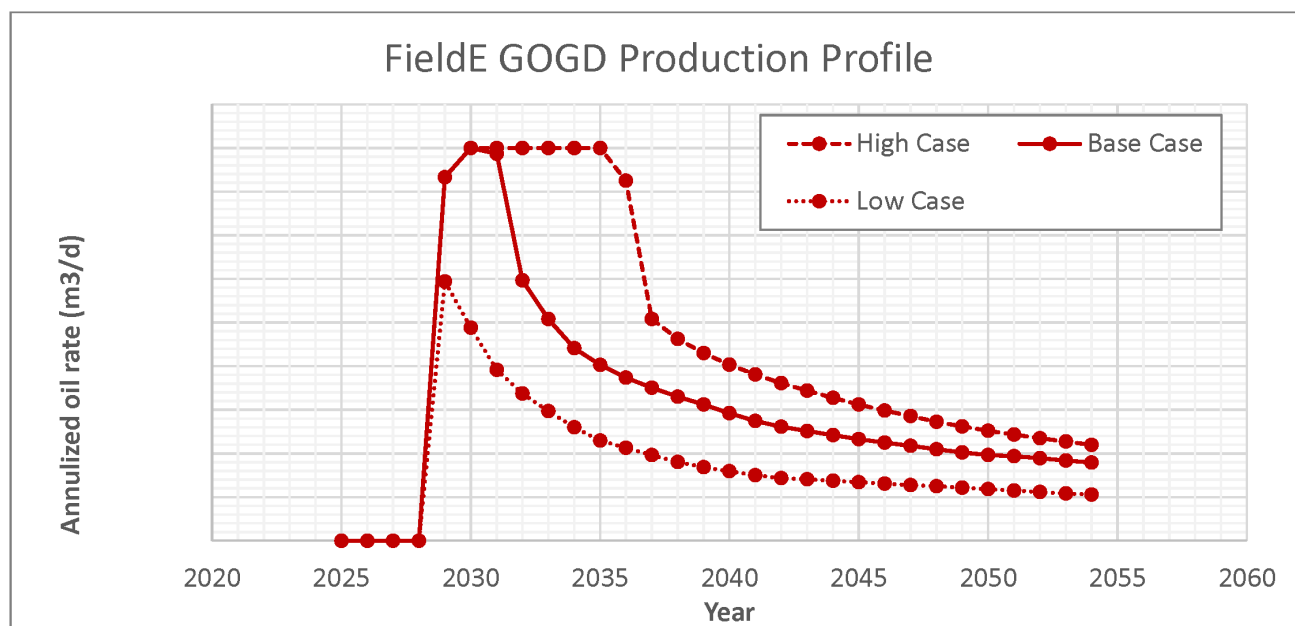


Figure 14—Low, base and high case oil recovery profiles

Economic Metrics

The economic performance of the post-GOGD development case demonstrates strong project returns and capital efficiency.

Table 4—Summary of economic performance under base-case development.

Metric	Post-GOGD (Base Case)
Unit Technical Cost UTC	Low
Net Present Value (NPV)	Positive NPV
Payback Period	Minimum

Several strategic enablers underpinned the project's improved economic attractiveness:

- **Elimination of gas sweetening and desalting**, enabled by full gas recycling and lean injection design,
- **Fit-for-purpose greenfield facility design**: Surface layout was optimized to minimize compressor count, simplify flow routing, and ensure efficient handling of lean gas injection and H₂S-laden fluids—reducing overall lifecycle costs.
- **Deferred drilling of APO wells**: Allows capital flexibility and phasing, minimizing upfront exposure while maintaining reservoir surveillance readiness.
- **Zero-flaring policy**, aligning with PDO's net-zero objectives and reducing OPEX exposure to gas treatment and emissions penalties.

Key Risks and Mitigations

The Field E cGOGD development plan introduces several subsurface and surface risks inherent to gas injection in fractured carbonate reservoirs. These risks were proactively addressed through an integrated risk assessment and mitigation framework, covering technical, operational, and economic dimensions. This section summarizes the top-ranked risks and the corresponding mitigation measures adopted in the field development plan.

Subsurface Risks

Risk	Impact	Mitigation Strategy
Early gas breakthrough	Reduces oil recovery, increases GOR, impacts compressor loading	• Crest injectors placed away from conductive fractures • Short perforation intervals • Use of Gas Breakout Control (GBC) technology
Fracture connectivity uncertainty	Affects sweep efficiency and breakthrough timing	• 7 fracture scenarios modeled • Conservative injector placement • Monitoring via observation wells
Poor conformance in GOGD	Limited vertical sweep, early cresting	• Injection rate optimization • Producers placed at mid-flank • GOC movement tracked via obs wells
Reservoir souring (H ₂ S evolution)	Health, safety, and material risk	• Souring studies conducted • Lean sweet injection gas selected • NACE-compliant material selection
Aquifer encroachment	Potential water production and handling cost	• No water injectors used • APO wells retained as contingency • Water cut monitored through phased startup

Surface and Facility Risks

Risk	Impact	Mitigation Strategy
Compressor limitations	Insufficient injection pressure or turndown control	• Brownfield booster compressors selected • Designed for 9,000 kPa delivery • Modular scalability built-in
Gas processing upsets	Risk of unplanned flaring or deferred injection	• Full gas recycling loop with redundant compression • H₂S monitoring with emergency shutdown system
Material integrity under H ₂ S	Risk of corrosion and equipment failure	• NACE MR0175 materials for all well and surface metallurgy
Delayed tie-in of surface facilities	Potential production delay and deferral of cash flow	• Surface-subsurface phasing aligned • Early long-lead items identified and tendered upfront

Economic and Project Delivery Risks

Risk	Impact	Mitigation Strategy
Oil price volatility	Reduced project NPV or delayed payback	- Robust base case economics - Short payback - Phase-wise investment plan enables agility
CAPEX overrun	Impact on UTC and project competitiveness	Fit-for-purpose greenfield facility designed with minimal compression and no gas sweetening or desalting
Execution delays	Loss of value from production deferral	- Execution readiness and material ordering aligned with phasing - Critical path items secured in early planning

Surveillance and Adaptive Management

To proactively manage uncertainty and optimize the development plan over time, the following surveillance and feedback loops were embedded:

- Observation wells for pressure and GOC tracking in both reservoirs.
- Production logging and GOR monitoring to detect early gas breakthrough.
- APO well conversion held in reserve if unexpected aquifer influx occurs.
- Integrated digital surveillance dashboards for injector-producer performance tracking.
- Technical limits (TL) tracking as part of ongoing project performance assurance

Conclusions and Recommendations

This paper has presented the successful planning, evaluation, and justification of a cold Gas-Oil Gravity Drainage (cGOGD) redevelopment in Field E, targeting two fractured, low-permeability carbonate reservoirs. The integrated strategy combines analogue- informed reservoir engineering, fit-for-purpose greenfield facilities, and robust risk mitigation, transforming a technically stranded opportunity into a high-value, executable development.

Key Conclusions

1. Gravity-driven recovery is effective in fractured carbonates:
Cold gas injection from crestal injectors, coupled with mid-flank horizontal producers, achieved a significant uplift in recovery factor—doubling historical RF to ~17-18% across both Natih E and Shuaiba.
2. Tailored development design was essential:
Dual-reservoir modeling, GOC-sensitive producer placement, and tight control over injector perforation depth enabled pressure support and delayed gas breakthrough, maximizing sweep efficiency.
3. Greenfield facility design unlocked feasibility:
Custom-built surface infrastructure with modular gas compression and lean gas handling, specifically optimized for cGOGD, allowed phasing flexibility, early production readiness, and long-term injectivity support.

4. Economic fundamentals are strong and resilient:
With a base-case Positive NPV, low Unit Technical Cost (UTC), and a lower payback period, the project delivers robust returns even under downside scenarios.
5. Risk mitigation is embedded across subsurface and surface design:
Strategies included fracture corridor avoidance, GBC (Gas Breakout Control) implementation, lean sweet injection gas, and H₂S-resistant materials—all contributing to technical assurance.
6. ESG alignment is demonstrated through zero-flaring:
The development meets PDO's emissions reduction targets through full gas reinjection, closed-loop compression, and sour service compliance—without requiring sweetening or water handling.

Recommendations for Similar Fields

- Start with analogue benchmarking:
Insights from nearby successful GOGD deployments guided key decisions on injector placement, APO phasing, and surveillance strategy.
- Ensure early subsurface-surface integration:
Aligning compression specs, injection pressure limits, and well pad layout from the outset reduced risk of rework and enabled an executable facilities concept.
- Model uncertainty, not just the base case:
Incorporating multiple fracture realizations, injector conformance sensitivities, and pressure behavior scenarios strengthened design resilience.
- Embed flexibility in development planning:
Retaining APO wells as contingency, sequencing drilling to match gas availability, and modular surface design allowed for phased de-risking and operational agility.

This cGOGD development in Field E provides a replicable model for maximizing recovery from mature, complex carbonate reservoirs—without large-scale waterflooding or stimulation. Through disciplined integration of engineering and execution, the project unlocks significant value and supports long-term sustainability objectives.

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